NGO & SONS
ACADEMIC COACHING

Name: _____

Class: _____

Date: _____

YEAR 7 ALGEBRA

Junior Mathematics

Lecture 1	1. Y7 Indices Exam (30 mins) 2. Using Pronumerals 3. Substitution 4. Algebraic Patterns
Lecture 2	1. Algebraic Patterns 2. Algebraic Operations
Lecture 3	1. Algebraic Operations 2. Advanced Algebra
Lecture 4	1. Fraction Algebra 2. Factorisation

Instructions

- For a thorough understanding of the topic, **every question in this handout is to be completed.**
- When the teacher is going through the example, **no writing is permitted.**
- **Do not skip ahead** of the teacher and complete the questions at the same pace as the teacher.
- **After copying the working out** on a separate piece of paper move onto the **tutorial booklet**.

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6. Fraction Algebra

To add or subtract fractions, you must have a common denominator.

When finding the common denomination, you must find the LCM between the bottom numbers.

Examples: Simplify the following.

1. $\frac{x}{5}+\frac{x}{3}$

Note:

2. $\frac{2x^2}{3}-\frac{5x^2}{6}$

Note:

3. $\frac{x+4}{3}+\frac{2x-1}{8}$

Note:

4. $\frac{2x-1}{6}-\frac{x+3}{4}$

Note:

5. $\frac{3}{x}+\frac{1}{2x}$

Note:

6. $\frac{5}{x^2}-\frac{4}{x}$

7. $\frac{1}{x^2y}+\frac{1}{xy^2}$

Note:

8. $\frac{2}{5a^2b}+\frac{3}{10ab}$

7. Factorisation

Factorisation is the opposite of expanding.

$$\xrightarrow{\textit{expanding}}$$

$$a(b+c)=ab+ac$$

$$\xrightarrow{\textit{factorising}}$$

$$ab+ac=a(b+c)$$

A) Simple Factorisation

Always ask:

1. "What's common?"
2. "What's left?"

When factorizing, look at:

1. Numbers first
2. Then, letters in alphabetical order – taking the lowest power
3. What's left – again looking at numbers first, then letters

Examples: Factorise the following.

1. $2a + 6$

2. $a^2 + 2a$

3. $8x^2 - 6x$

Note:

4. $2a^3b^2 + 4ab^2$

5. $16a^4bc - 12a^3b^2$

B) Factorising out the negative

When dealing with negative terms, you must think about the following rules:

1. If the first term is negative, factorise out the (-) sign as well.
2. When taking out the (-) sign, it flips all the signs of the terms.

E.g. $-x - 3 = -(x + 3)$

Examples: Factorise the following.

1. $-5a - 26$

2. $-4x - 8$

3. $-4x^3 - 2x$

4. $-5m^2p + mp^3$

C) Complicated Factorisation

Whenever two terms have a common bracket, the entire bracket can be taken out.

You must ask:

1. "What's common"
2. "What's left"

Examples: Factorise the following.

1. $4(3x - 2) + x(3x - 2)$

Note:

2. $2(x + 1) - x(x + 1)$

Note:

3. $x(x^2 + 1) + (x^2 + 1)$

Note:

4. $2x(x^2 - 1) - (x^2 - 1)$

D) Simplifying Fractions with Brackets

Simplifying fractions with brackets

$$\frac{2(3)}{2} = 3$$

$$\frac{2 + 3}{3} \neq 3$$

$$\frac{2(x + 1)}{2} = x + 1$$

$$\frac{2 + (x + 1)}{2} \neq x + 1$$

Can only simplify fractions if there are multiplication of terms on the top AND bottom.

Sometimes need to factorise to get a multiplication of terms

E.g. $2x + 2 = 2(x + 1)$

Examples: Simplify the following.

1. $\frac{2x+4}{6}$

2. $\frac{x^2-3x}{3x}$

3. $\frac{3x^2-6x}{9x^2}$

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YEAR ALGEBRA

Junior Mathematics

Tutorial 4	1. Fraction Algebra 2. Factorisation

Instructions

- For a thorough understanding of the topic, **every question in this handout is to be completed.**
- **Attempt** the following questions **after** copying the examples and during tutorials.
- If you are **stuck on a question,** ask the teacher for help during tutorials.
- **Tutorial booklet** is to be checked off by teacher at the end of the lesson.

6. Fraction Algebra

Classwork: Simplify the following.

1. $\frac{2x}{5} - \frac{x}{3}$
2. $\frac{1}{3}y - \frac{5}{12}y$
3. $\frac{4}{5}ab - \frac{1}{3}ab$
4. $\frac{x+3}{4} + \frac{2x-1}{3}$
5. $\frac{6x+1}{12} + \frac{2x+3}{8}$
6. $\frac{2x+1}{2} - \frac{x+1}{3}$
7. $\frac{3x-1}{9} - \frac{4-x}{6}$
8. $\frac{3}{2y} - \frac{4}{y}$
9. $\frac{b}{a^3} - \frac{2}{a^2}$
10. $\frac{3}{x^3y^2} - \frac{2}{x^2y^3}$
11. $\frac{4}{5x^2y^2} - \frac{3}{10xy^3}$
12. $\frac{4}{15a^2bc^3} - \frac{1}{6ab^2c^3}$
13. $x^2(1+x) + (1+x)$
14. $2x(3x+1) + (3x+1)$
15. $\frac{3x+18}{9}$
16. $\frac{10x-15}{5}$
17. $\frac{x}{2x^2+3x}$
18. $\frac{4x-6x^2}{6x^2}$
19. $\frac{x^2+xy}{xy}$

7. Factorisation

Classwork: Factorise the following.

1. $9a - 27$
2. $5a^2 + a^3$
3. $2x^3y + 3xy^2$
4. $4c^2d^3 - 6cd^3$
5. $12ab^2c - 21a^2b^2c^2$
6. $-5a - 10$
7. $-5x^2 + x^3$
8. $-2xy + 4x^2y$
9. $-6a^2b^3 - 4ab^3c$
10. $-12mn^3l^2 - 18m^2n^2l$
11. $3(x+1) + (x+1)$
12. $4(1-x) - x(1-x)$

Name: _____

Class: _____

E.g. Y8 Sat 12:20-4:05 PARRA

Teacher: _____

Term: _____ Week: _____

YEAR 7 ALGEBRA

Homework – Lecture 4

Instructions

- **Attach this page to the front** of your homework when submitting.
- **Write out the full question** in your homework unless instructed otherwise by the teacher.
- After completing the homework, use the answers at the end to **mark your work**.

Comments

Mark /10

Developmental Questions

1. Simplify the following.

a. $\frac{2x}{3} - \frac{x}{4} = \frac{4x}{12} - \frac{3x}{12} = \frac{x}{12}$

b. $\frac{4x}{5} + \frac{x}{15} = \frac{12x}{15} + \frac{x}{15} = \frac{13x}{15}$

c. $\frac{3xy}{4} - \frac{xy}{6} = \frac{9xy}{12} - \frac{2xy}{12} = \frac{7xy}{12}$

d. $\frac{3}{16}ab^2 - \frac{5}{12}ab^2 = \frac{9}{48}ab^2 - \frac{20}{48}ab^2 = \frac{-11}{48}$

e. $\frac{3}{16}ab^2 - \frac{5}{12}ab^2 = \frac{-11}{48}$

f. $\frac{x+3}{5} + \frac{x+2}{2} = \frac{2x+6}{10} + \frac{5x+10}{10} = \frac{7x+16}{10}$

g. $\frac{x-2}{3} + \frac{2x-1}{9} = \frac{3x-6}{9} + \frac{2x-1}{9} = \frac{5x-7}{9}$

h. $\frac{3x+1}{15} - \frac{2x+1}{10} = \frac{6x+2}{30} - \frac{6x+3}{30} = \frac{-1}{30}$

i. $\frac{6x-1}{3} - \frac{x+1}{4} = \frac{16x-4}{12} - \frac{3x+3}{12} = \frac{13x-7}{12}$

j. $\frac{2x-1}{8} - \frac{x-2}{12}$ $\frac{6x-3}{24} - \frac{2x-4}{24} = \frac{4x+1}{24}$

k. $\frac{2x-1}{8} - \frac{x-2}{12} = \frac{6x-3}{24} - \frac{2x-4}{24} = \frac{4x+1}{24}$

l. $\frac{1-x}{5} + \frac{3-2x}{3} = \frac{3-3x}{15} + \frac{15-10x}{15} = \frac{-12-7x}{15}$

m. $\frac{4x+1}{3} - \frac{2x-1}{4} = \frac{16x+4}{12} - \frac{6x-3}{12} = \frac{10x+7}{12}$

n. $\frac{2x-1}{3} + \frac{2-x}{6} = \frac{4x-2}{6} + \frac{2-x}{6} = \frac{3x}{6}$

o. $\frac{2x+1}{12} - \frac{1-x}{18} = \frac{6x+3}{36} - \frac{2-2x}{36} = \frac{4x+1}{36}$ Boring

p. $\frac{3}{4a} - \frac{3}{8a} = \frac{6}{8a} - \frac{3}{8a} = \frac{3}{8a}$ Lovely... Beans are healthy Free

q. $\frac{3}{4a} - \frac{3}{8a} = \frac{3}{8a}$

r. $\frac{3}{5a^2b} + \frac{1}{15ab} = \frac{3}{15a^2b} + \frac{3a}{15a^2b} = \frac{9a}{15a^2b}$

s. $\frac{4}{x^2yz} + \frac{2}{xy^2z} = \frac{?}{x^2y^2z} = \frac{4y}{x^2y^2z} + \frac{2x}{x^2y^2z} = \frac{4y+2x}{x^2y^2z}$

t. $\frac{a+1}{6ab^2} - \frac{1}{9a^2b} = \frac{?}{18a^2b^2} \Rightarrow \frac{3a^2+3a}{18a^2b^2} - \frac{2b}{18a^2b^2} = \frac{3a^2+3a-2b}{18a^2b^2}$

2. Simplify:

$$\frac{1-a}{3} - \frac{2-a}{5}$$

3. Simplify:

$$\frac{2m}{5} - \frac{m}{2} + \frac{m}{3}.$$

4. Simplify:

$$x - \frac{1}{y}..$$

5. Simplify the following.

a. $5x + 10$
b. $3x - 3y$
c. $10k + 12$
d. $3a^2 + 9a$
e. $15p^2q - 20pr$
f. $-6 + 12q$
g. $-uv - uw$
h. $-2a^2 - a^3$
i. $x(x+4) - 2(x+4)$
j. $-16u^2v - 24uw$
k. $ab(a-7) - (a-7)$
l. $x^2(x-1) + (x-1)$
m. $x(1-2x) + (1-2x)$
n. $4(2x-3) + x(2x-3)$
o. $\frac{20x-16y}{4}$
p. $\frac{8-8m}{16}$
q. $\frac{10u-5u^2}{2u}$
r. $\frac{24-32x}{16}$

Extension Questions:

6. Simplify the following expression

$$\left(a + \frac{3a}{1}\right)\left(a + \frac{5a}{4}\right)\left(a + \frac{7a}{9}\right)\ldots\left(+\frac{41a}{400}\right)$$

Answers:

1.

a. $\frac{5x}{12}$
b. $\frac{13x}{15}$
c. $\frac{7xy}{12}$
d. $-\frac{11ab^2}{48}$
e. $\frac{5x^2y^3}{9}$
f. $\frac{7x+16}{10}$
g. $\frac{5x-7}{9}$
h. $-\frac{1}{30}$
i. $\frac{21x-7}{12}$
j. $\frac{4x+1}{24}$
k. $\frac{29-2x}{20}$
l. $\frac{18-13x}{15}$
m. $\frac{10x+7}{12}$
n. $\frac{x}{2}$
o. $\frac{8x+1}{36}$
p. $\frac{3}{8a}$
q. $\frac{25-2a}{10a^2}$
r. $\frac{a+9}{15a^2b}$
s. $\frac{2x+4y}{x^2y^2z}$
t. $\frac{3a^2+3a-2b}{18a^22b^2}$

2. $\frac{-1-2a}{15}$

3. $\frac{7m}{30}$

4. $\frac{xy-1}{y}$

5.

a. $5(x+2)$
b. $3(x-y)$
c. $2(5k+6)$
d. $5p(3pq-4r)$
e. $-u(v+w)$
f. $(x+4)(x-2)$
g. $(a-7)(ab-1)$
h. $(1-2x)(x+1)$
i. $5x-4y$
j. $\frac{5(2-u)}{u}$
k. $3a(a+3)$
l. $-6(1-2q)$
m. $-a^2(a+2)$
n. $-8u(2uv+3w)$
o. $(x-1)(x^2+1)$
p. $(2x-3)(x+4)$
q. $\frac{1-m}{2}$
r. $\frac{3-4x}{2}$

6. $441a^{20}$